

Executive Summary

- **Signal:** The quarter-over-quarter change in the number of distinct 13F filers holding a stock, ranked across the S&P 500 (Ownership Breadth Momentum). The core idea relies on the link between growing ownership breadth and positive future stock returns: a growing holder base creates sustained upward price pressure, while a sharp drop signals eroding institutional support
- **Headline Result (60-day lag, 10bps cost):** The long-short spread yields a weak 2.65% annualized return and a 0.15 Sharpe ratio, backtested from 3Q15 to 1Q26. It severely underperforms both the SPY (14.99% return / 0.84 Sharpe) and an internal equal-weight benchmark. It also suffered a massive **-55.4%** Max Drawdown compared to SPY's -33.7%.
- **Key Offenders:** First, excessive turnover (~77-82% per quarter) destroys the strategy's edge once trading costs exceed 10bps. Second, the raw signal naturally leans toward shorting mega-cap stocks. This structural short bias dragged down performance during a decade-long bull market driven by these exact same companies.
- **Verdict:** As it stands, the strategy is a bust. The massive drawdown and high turnover make it completely untradable. Although the Rank-IC (around 0.02) is consistently positive across all lags, it's drowned out by noise (t-stats ~1.0).

Factor Hypothesis

$$Signal_{i,t} = Rank(Breadth_{i,t} - Breadth_{i,t-1})$$

$Breadth_{i,t}$ is the count of distinct institutional investors (CIKs) reporting a position in a stock. To ensure strict point-in-time compliance, both the current and prior quarter breadth are evaluated using only the filings publicly available as of the current formation date.

The core intuition is simple: we are tracking the expansion or contraction of institutional consensus. When a stock is held by very few funds, its price is dictated by a narrow group of optimists. When the number of holders increases, it means a broader, independent set of institutional players are validating the stock and bringing fresh capital. Conversely, a shrinking holder base means institutions are quietly heading for the exits.

We specifically chose this "headcount" approach over two obvious alternatives:

1. Value-weighted "smart money" following (sizing by reported dollar value) : Counting total dollars conflates a single massive fund making a concentrated bet with a broad market consensus. One fund buying \$1B is a single opinion; 100 different funds buying \$10M each represents a strong, distributed trend. We are trading the consensus, not the concentration of capital.
2. Only "Smart Money" Following (tracking or heavily weighting the signal based only on the buys and sells of top-tier, famous hedge funds): Weighting the signal by top-tier funds introduces severe hindsight bias (we know today who the best funds of the last decade were). It also conflates a concentrated bet by a single manager with genuine market-wide validation.

Scope Decisions

- **Holder universe:** All 13F filers, minus a hardcoded exclusion list of major passive indexers (Vanguard, BlackRock, State Street, Geode, Northern Trust, Schwab). Index tracking is mechanical flow, not an active positioning signal. Hardcoding this list was a necessary pragmatic choice for the research phase, but it is a known technical limitation. A production pipeline should replace this static list with an automated CIK-level classification engine to dynamically separate passive index arms from active fundamental funds.
- **Equity universe:** S&P 500, point-in-time (not today's constituent list, to avoid Survivorship Bias) — bounds the CUSIP-mapping problem and the price-history requirement to a tractable, well-covered set.
- **Time Window :** 2015-2026, large period to increase the number of independent quarterly observations, giving our statistical significance tests (t-stats) an honest, adequately sized sample.

Deliberate Engineering Exclusions

- **Full CUSIP/Ticker Genealogy:** Model relied on Current-Ticker resolution using the OpenFIGI API for mapping data, supplemented by four manual overrides for major corporate actions. Building a complete historical mapping engine from scratch was out of scope and unlikely to change the core statistical outcome.
- **Complex Amendment Parsing:** Every 13F-HR/A (amendment) is treated as a full replacement of the original filing. The SEC's "*AMENDMENTTYPE*" flags are notoriously inconsistent in practice, making surgical merge rules too dangerous to trust.
- **Value-Weighting:** Both the signal and the portfolio construction are strictly equal-weighted. Calculating the signal using the raw change in holder count naturally pushes mega-cap stocks into the extreme ranks (Deciles 1 and 10) simply because their baseline holder count is massive (e.g., losing a mere 5% of 2,000 holders is a larger absolute drop than losing 20% of 200 holders). Because the signal inherently carries this mega-cap tilt, applying market-cap weighting to the portfolio on top of it would turn the strategy into a concentrated bet on three or four tech giants.
- **Confidential Treatment Requests (CTRs):** Positions legally omitted by managers under SEC confidential treatment are ignored. They are fundamentally unrecoverable from public point-in-time data.
- **Survivorship bias free source of data:** Historical closing price data were retrieved using the *yfinance* library for simplicity. However, *yfinance* is subject to survivorship bias, as it primarily provides data for currently active tickers, excluding securities that have been delisted or otherwise ceased trading.

Methodology

The **data pipeline ingests the SEC’s bulk** structured datasets. It parses and relationalizes the records, specifically **collapsing split voting authorities** within filings to ensure each institution is counted exactly once. This processed data is compiled into a single Parquet file for data efficiency.

Point-in-time discipline is handled once, at the lowest layer, and every downstream stage inherits it. To avoid lookahead bias, the model evaluates history purely through time-stamped snapshots. On rebalance date, the **system queries for the most recently accepted submission on or before that exact timestamp**. If a fund delays its filing past the rebalance window, the model treats that manager as absent, never retroactively backfilling records when an amendment or late submission arrives months later. This forces the backtest to operate within the bounds of real-time visibility.

While SEC filings are due 45 days post-quarter-end, historically only 87.0% arrive by that deadline, increasing to 96.3% by day 60 and 96.8% by day 90. To avoid cherry-picking an optimal timeframe (data snooping), **the backtest sweeps 45, 60, and 90-day formation lags as three independent runs**.

The **portfolio rebalances quarterly, sorting equities into signal deciles to form an equal-weighted, dollar-neutral long-short structure** (long top decile, short bottom decile) that isolates pure factor returns from market beta. Positions are marked to market daily using *yfinance* adjusted close prices, capturing true daily volatility and drawdown metrics rather than smoothed quarterly estimates.

Transaction costs are evaluated across a one-way grid of 0, 5, 10, and 25 basis points. The full round-trip execution penalty is calculated by doubling the total portfolio turnover (long and short legs) at each quarterly rebalance. Because this friction is incurred at execution, the penalty is applied as a single multiplicative haircut to the quarter’s daily NAV curve, mirroring the reality of paying execution costs on day one and compounding the net capital forward.

Performance is evaluated against both SPY and an internal equal-weight universe, which controls for potential sampling bias within our mappable subset of the S&P 500. Data integrity requires that assets with missing daily pricing are dropped, not imputed, in order to avoid misleading our factor.

Results

The 10.5-year backtest (Q3 2015–Q1 2026) processed 81.8M rows, evaluating all 12 lag/cost scenarios across ~42 quarterly rebalances. Of 515 total tickers, just 4 were dropped for missing prices, proving the CUSIP-mapping gap negligible. The scored cross-section maintained high liquidity, averaging 370 tradable names per quarter (range: 304–428).

The results are objectively weak. The strategy is barely profitable, exhibiting market-level volatility without the corresponding returns and a drawdown nearly double SPY’s (as seen in the graph). It underperforms the internal equal-weight benchmark across all risk-adjusted metrics, proving it non-viable as a standalone fundable strategy.

Primary result (60-day lag, 10bps):

	Long-short spread	Internal universe (EW)	SPY
Annualized return	2.65%	12.96%	14.99%
Annualized volatility	17.19%	18.17%	17.85%
Sharpe ratio (rf = 0)	0.15	0.71	0.84
Maximum drawdown	-55.4%	-41.2%	-33.7%
Hit rate (daily)	52.8%	54.6%	55.4%

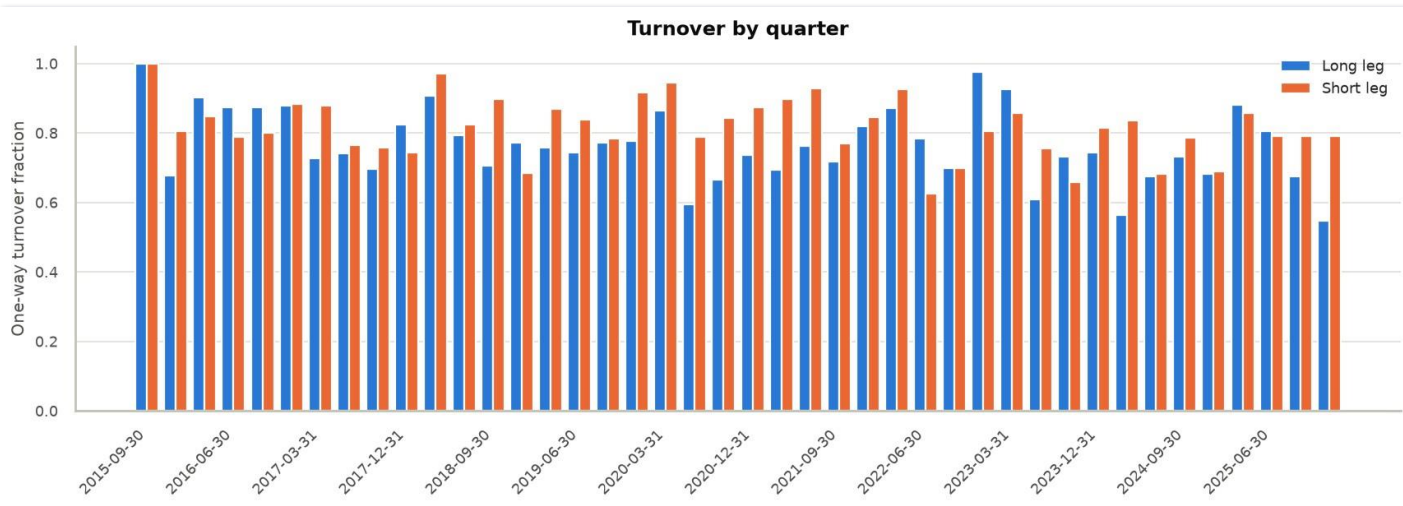
Primary backtest: long-short spread vs. benchmarks



Lag sensitivity is non-monotonic and does not confirm a front-loaded-decay mechanism. At 10bps: 45- day Sharpe 0.36 (max DD -28.4%), 60-day Sharpe 0.15 (max DD -55.4%), 90-day Sharpe 0.26 (max DD -43.5%). The sweep is inconclusive on decay, most consistent with lag-to-lag variation being dominated by noise at this sample size (42-43 observations per lag) rather than a clean information-decay curve.

Lag sensitivity (cost = 10bps)





- **High Turnover & Friction:** high churn (~80% quarterly replacement per leg) near decile boundaries severely degrades returns.
- **Statistical Significance:** Rank-IC is weakly positive (0.016 to 0.023) but statistically insignificant (t-stats of 0.6–1.1) due to extreme quarterly volatility.

Next Steps

In its current form, the strategy is not viable for live capital deployment and requires substantial redevelopment before reaching production readiness. With that said, here are the next three steps to improve the research:

1. Deep-Dive Backtest Analysis & Edge Verification: As this initial work only provides a high-level baseline, a comprehensive performance evaluation has not yet been performed. A deep-dive analysis is urgently required to thoroughly investigate the backtest results—specifically comparing long versus short leg performance, assessing behavior across different macro regimes, and isolating true alpha from systematic factor exposures. Completing this deep analysis is necessary to prove whether a genuine, tradeable edge exists before deciding to allocate further resources to this model.

2. Reduce Trading Turnover: Constant buying and selling—caused by stocks bouncing in and out of the top rankings—is eating up profits. Future versions should add a "buffer zone" (so we don't sell a stock just because its rank drops slightly) or smooth the data over time to ignore random noise and only trade on real, lasting trends.

3. Refined Breadth Momentum Scaling (Signal Engineering): The current ownership breadth momentum signal is inherently biased: nominal changes overweight mega-cap equities, while percentage changes overweight micro-caps. Developing a cross-sectionally normalized metric—such as scaling by total institutional float, adjusting for market cap, or using regression residuals—will provide a cleaner, unbiased signal.